

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCVE705D - ROCK MECHANICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 The primary objective of rock mechanics in civil engineering is to
 - A. study mineral composition of rocks
 - B. analyze behavior of rock under applied loads
 - C. determine age of rock formations
 - D. classify rocks based on origin
 - 2 Rock mechanics finds *least* application in
 - A. slope stability analysis
 - B. underground excavations
 - C. rock support system design
 - D. irrigation water quality analysis
 - 3 Strength and deformation characteristics of rocks are *not* influenced by
 - A. confining pressure
 - B. loading rate
 - C. moisture condition
 - D. geographical latitude
 - 4 Uniaxial compressive strength test is primarily used to determine
 - A. shear strength of intact rock
 - B. tensile strength of rock mass
 - C. compressive strength of rock specimen
 - D. permeability of rock
 - 5 Slake durability test is used to assess
 - A. resistance of rock to chemical attack
 - B. weathering resistance under wetting–drying cycles
 - C. permeability characteristics
 - D. elastic behavior of rocks
 - 6 Rock Quality Designation (RQD) is based on
 - A. rock core recovery percentage
 - B. length of sound core pieces
 - C. rock density variation
 - D. number of joints per meter

- 7 Geological Strength Index (GSI) primarily considers
 - A. intact rock strength only
 - B. joint orientation and spacing
 - C. rock mass structure and surface condition
 - D. depth of rock formation
 - 8 Rock Mass Rating (RMR) system includes all except
 - A. UCS of intact rock
 - B. groundwater conditions
 - C. rock color
 - D. joint condition
 - 9 Plane failure in rock slopes occurs mainly due to
 - A. circular slip surfaces
 - B. weak bedding planes dipping out of slope
 - C. weathered soil cover
 - D. high pore water pressure only
 - 10 Wedge failure is commonly associated with
 - A. single joint set
 - B. random fractures
 - C. intersection of two joint planes
 - D. weak intact rock
 - 11 Mohr–Coulomb failure criterion is based on
 - A. tensile strength only
 - B. shear strength parameters
 - C. elastic constants
 - D. permeability factors
 - 12 Shotcrete is primarily used in rock engineering to
 - A. improve tensile strength of rock
 - B. prevent weathering and provide surface support
 - C. increase permeability
 - D. reduce joint spacing
- Q2. Solve the following.
- A) State and explain the principles and scope of rock mechanics. 6
 - B) Write a short note on factors influencing deformation of rocks. 6
- Q3. Solve the following.
- A) Explain the procedures and equipment used in core drilling for rock sampling. 6
 - B) Describe the triaxial compressive strength test and its applications. 6
- Q4. Solve Any Two of the following.
- A) Explain the procedure for determining Rock Quality Designation (RQD). 6
 - B) Explain NGI (Q-system) classification and its applications. 6
 - C) Explain the Geological Strength Index (GSI) and its significance. 6
- Q5. Solve Any Two of the following.
- A) Explain the modes of failure in rock slopes with neat sketches. 6
 - B) Describe methods of monitoring rock slope failure. 6
 - C) Explain wedge failure and its conditions of occurrence. 6

Q6. Solve Any Two of the following.

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|---|---|
| A) Describe the Mohr–Coulomb failure criterion and its application. | 6 |
| B) Explain rock bolting and shotcrete as support systems. | 6 |
| C) Describe rock reinforcement techniques. | 6 |

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